

Fractional Brownian Motion: Small Increments and First Exit Time from One-sided Barrier

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May 22, 2024



- The fBm was first introduced by Kolmogorov in 1940 and popularized by Mandelbrot & van Ness in 1968.
- The fBm $\{B_H(t)\}_{t \geq 0}$ with Hurst parameter H is the unique zero-mean continuous Gaussian process with $B_H(0) = 0$ a.s. and whose covariance function is given as:

$$\mathbb{C}ov(B_H(s), B_H(t)) = \frac{C_H}{2} (|s|^{2H} + |t|^{2H} - |s - t|^{2H}),$$

where $C_H = \mathbb{V}ar(B_H(1)) > 0$ only depends on H .

- When $H = 1/2$, fBm extends Bm.

- (Statistical self-similarity) $\{B_H(t)\}_{t \geq 0}$ is H -self-similar: for all $a > 0$,

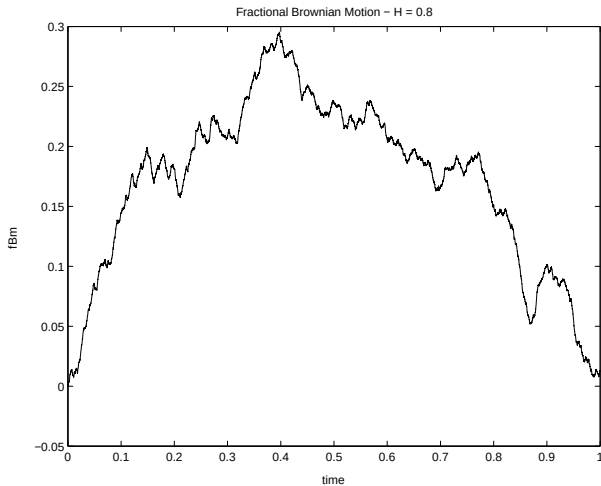
$$\{B_H(at)\}_{t \geq 0} \stackrel{\text{law}}{=} \{a^H B_H(t)\}_{t \geq 0}.$$

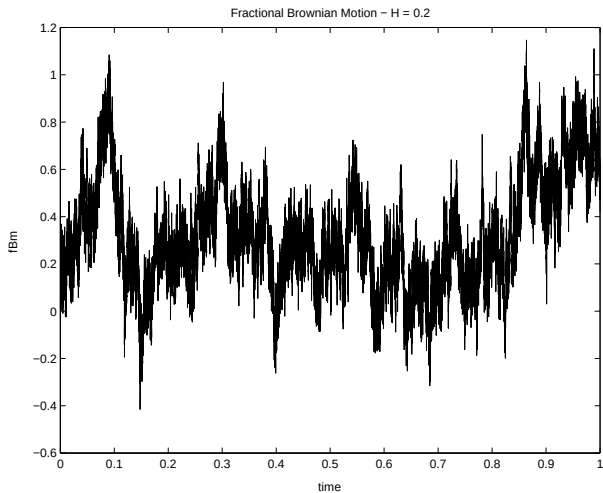
- (Stationary increments) $\{B_H(t)\}_{t \geq 0}$ has stationary increments: for all $s > 0$,

$$\{B_H(s+t) - B_H(s)\}_{t \geq 0} \stackrel{\text{law}}{=} \{B_H(t) - B_H(0)\}_{t \geq 0}.$$

- (Increments are generally dependent) For $t > s$,
 - when $H > 1/2$, $\text{Cov}(B_H(t) - B_H(s), B_H(s)) > 0$; (long-memory property)
 - when $H < 1/2$, $\text{Cov}(B_H(t) - B_H(s), B_H(s)) < 0$;
 - when $H = 1/2$, $\text{Cov}(B_H(t) - B_H(s), B_H(s)) = 0$. (independent increments)

Rough Paths of FBm





Let τ_b denote the first exit time of B^H from the upper barrier $b > 0$, then we define the upper-tail probability of τ_b as: for any $T \geq 0$,

$$\mathbb{P}(\tau_b > T) = \mathbb{P}\left(\sup_{t \in [0, T]} B^H(t) \leq b\right). \quad (1)$$

- Asymptotic behavior as $T \rightarrow +\infty$: by the self-similarity,

$$\begin{aligned} \mathbb{P}\left(\sup_{t \in [0, T]} B^H(t) \leq b\right) &= \mathbb{P}\left(\sup_{t \in [0, 1]} B^H(tT) \leq b\right) \\ &= \mathbb{P}\left(\sup_{t \in [0, 1]} B^H(t) \leq bT^{-2H}\right) \xrightarrow{T \rightarrow +\infty} 0. \end{aligned}$$

- Asymptotic rate: How fast is the above convergence?

- For $B^{1/2}$ (Bachelier, 1900): there exist $c_1, c_2 > 0$ such that

$$c_1 T^{-1/2} \leq \mathbb{P}(\tau_b > T) \leq c_2 T^{-1/2}, \text{ as } T \rightarrow +\infty.$$

Proof. For B_m , τ_b has a Lévy distribution density

$$f_{\tau_b}(x) = \begin{cases} \frac{b}{\sqrt{2\pi x^3}} \exp\left(-\frac{b^2}{2x}\right) & \text{for } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

- Conjecture: for B^H , is it true that

$$c_1 T^{-(1-H)} \leq \mathbb{P}(\tau_b > T) \leq c_2 T^{-(1-H)}, \text{ as } T \rightarrow +\infty? \quad (2)$$

- Our dream is to prove the conjecture (2).

Why it is so hard to prove the conjecture?

Assume we were undergraduate Maths students: We believe, for n large enough,

$$\begin{aligned}\mathbb{P}\left(\sup_{t \in [0, T]} B^H(t) \leq b\right) &\approx \underbrace{\mathbb{P}\left(B^H(t_1) \leq b, \dots, B^H(t_n) \leq b\right)}_{0 < t_1 < t_2 < \dots < t_n = T} \\ &= \int_{(-\infty, b]^n} f_{(B^H(t_1), \dots, B^H(t_n))}(x_1, \dots, x_n) dx_1 \dots dx_n.\end{aligned}$$

We then realized: Proving the conjecture requires a deep understanding of the multiple integral

$$\int_{-\infty \leq x_1, \dots, x_n \leq b} f_Z(x_1, \dots, x_n) dx_1 \dots dx_n, \quad (3)$$

where f_Z is the density function of the Gaussian vector $Z \sim \mathcal{N}(\mu, \Sigma)$, with any μ, Σ .

However: Explicit solutions or even good approximations of (3) have not been yet found. This problem requires a deep understanding of the behavior of Σ^{-1} , which is a classical but challenging problem in linear algebra.....

Multivariate normal density functions

For $Z \sim \mathcal{N}(\mu, \Sigma)$:

Matrix representation:

$$f_Z(x) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right\}.$$

2-d expression:

$$f_Z(x, y) = \frac{1}{2\pi \sigma_X \sigma_Y \sqrt{1 - \rho^2}} \exp \left(-\frac{\left(\frac{x - \mu_X}{\sigma_X}\right)^2 - 2\rho \left(\frac{x - \mu_X}{\sigma_X}\right) \left(\frac{y - \mu_Y}{\sigma_Y}\right) + \left(\frac{y - \mu_Y}{\sigma_Y}\right)^2}{2(1 - \rho^2)} \right).$$

3-d expression (by ChatGPT):

$$|\Sigma| = \sigma_{xx}(\sigma_{yy}\sigma_{zz} - \sigma_{yz}^2) - \sigma_{xy}(\sigma_{xy}\sigma_{zz} - \sigma_{yz}\sigma_{xz}) + \sigma_{xz}(\sigma_{xy}\sigma_{yz} - \sigma_{yy}\sigma_{xz});$$

$$\Sigma^{-1} = \frac{1}{|\Sigma|} \begin{pmatrix} \sigma_{yy}\sigma_{zz} - \sigma_{yz}^2 & \sigma_{xz}\sigma_{yz} - \sigma_{xy}\sigma_{zz} & \sigma_{xy}\sigma_{yz} - \sigma_{xz}\sigma_{yy} \\ \sigma_{xz}\sigma_{yz} - \sigma_{xy}\sigma_{zz} & \sigma_{xx}\sigma_{zz} - \sigma_{xz}^2 & \sigma_{xz}\sigma_{xy} - \sigma_{xx}\sigma_{yz} \\ \sigma_{xy}\sigma_{yz} - \sigma_{xz}\sigma_{yy} & \sigma_{xz}\sigma_{xy} - \sigma_{xx}\sigma_{yz} & \sigma_{xx}\sigma_{yy} - \sigma_{xy}^2 \end{pmatrix}.$$

n -d expression: **Hopeless!**

- Conjecture: for B^H , there exist $c_1, c_2 > 0$ such that

$$c_1 T^{-(1-H)} \leq \mathbb{P}(\tau_b > T) \leq c_2 T^{-(1-H)}, \text{ as } T \rightarrow +\infty?$$

- (Molchan, 1999): there is $c > 0$ such that

$$T^{-(1-H)} e^{-c\sqrt{\log T}} \leq \mathbb{P}(\tau_b > T) \leq T^{-(1-H)} e^{c\sqrt{\log T}}, \text{ as } T \rightarrow +\infty. \quad (4)$$

- (Aurzada, 2011): for $\gamma_1, \gamma_2 > 0$ arbitrarily small,

$$T^{-(1-H)} (\log T)^{-1/(2H)-\gamma_1} \leq \mathbb{P}(\tau_1 > T) \leq T^{-(1-H)} (\log T)^{2/H-1+\gamma_2}, \text{ as } T \rightarrow +\infty. \quad (5)$$

- (Aurzada et al., 2018): for $H > 1/2$ (long-memory), there is $c > 0$ arbitrarily small,

$$c T^{-(1-H)} (\log T)^{-1/(2H)} \leq \mathbb{P}(\tau_1 > T) \leq c T^{-(1-H)}, \text{ as } T \rightarrow +\infty. \quad (6)$$

We are awaiting with eagerness Aurzada's updates on the conjecture, but it seems that the author turns to look at persistence probability problems of other processes....

- θ is called the persistence exponent if $\mathbb{P}(\tau_b > T) = T^{-\theta+o(1)}$ as $T \rightarrow +\infty$.
- For B^H , $\mathbb{P}(\tau_b > T)$ has persistence exponent $\theta = 1 - H$.
- But it is not over yet. The goal of this framework is to illuminate a direction of research that aims at improving the accuracy of estimation of the loss factor $T^{o(1)}$.

Theorem (Peng and Rao, 2023)

Let the integer $N \geq 6$, which can be arbitrarily large. There are two constants $c_1, c_2 > 0$ such that for any $\gamma > 0$ arbitrarily small,

$$\mathcal{L}(T, H, c_1) \leq \mathbb{P}(\tau_1 > T) \leq \mathcal{U}(T, H, N, c_2, \gamma), \text{ as } T \rightarrow +\infty, \quad (7)$$

where the asymptotic lower bound

$$\begin{aligned} & \mathcal{L}(T, H, c) \\ &= \begin{cases} cT^{-(1-H)}(\log T)^{-1/(2H)}(\log_2 T)^{-1/(2H)}(\log_3 T)^{-1/(2H)} & \text{if } H \in (0, 1/2); \\ cT^{-(1-H)}(\log T)^{-1/(2H)} & \text{if } H \in (1/2, 1) \end{cases} \end{aligned} \quad (8)$$

and the asymptotic upper bound

$$\begin{aligned} & \mathcal{U}(T, H, N, c, \gamma) \\ &= \begin{cases} T^{-(1-H)}(\log T)^{1/H-1}(\log_2 T)^{3/(2H)} \\ \quad \times ((\log_3 T)(\log_4 T))^{1/H}(\log_5 T)^{1/(2H)}(\log_N T)^\gamma & \text{if } H \in (0, 1/2); \\ cT^{-(1-H)} & \text{if } H \in (1/2, 1). \end{cases} \end{aligned} \quad (9)$$

Why are we able to improve?

Aurzada (2011) used $|B^H(s) - B^H(t)| \leq S|t - s|^{H-\gamma}$, while we use **an improved version**:

Theorem (Small increments of fBm, Peng and Rao, 2023)

Let $\{B^H(t)\}_{t \in [0,1]}$ be an fBm with Hurst parameter $H \in (0, 1)$. There exists a random variable $C > 0$ having finite moment of any positive order, such that for every $t \in [0, 1]$ and any $|h| \in (0, \min\{1 - t, e^{-e}\}]$,

$$|B^H(t+h) - B^H(t)| \leq C|h|^H \sqrt{\log \log |h|^{-1}}, \text{ a.s..} \quad (10)$$

Moreover, there is a constant $c > 0$, such that for any order $p > 0$,

$$\mathbb{E}[C^p] \leq c^p \sqrt{\Gamma\left(p + \frac{1}{2}\right)}, \quad (11)$$

where Γ is the gamma function.

Idea of the proof of the small increments of fBm theorem:

- The proof relies on an optimal rate random wavelet series representation of B^H (Meyer, Sellan and Taqqu, 1999): for any $t \in [0, 1]$,

$$B^H(t) = \sum_{j=-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} 2^{-jH} (\Psi_H(2^j t - k) - \Psi_H(-k)) \varepsilon_{j,k}, \text{ a.s.},$$

where Ψ_H are carefully chosen wavelet functions and $\varepsilon_{j,k}$ are i.i.d. standard normal variables.

- The existence of C and the upper bound of $\mathbb{E}[C^p]$ can be obtained following Ayache and Taqqu (2003).

Corollary (Peng and Rao, 2023)

Let $T > 0$. There exists a random variable $C > 0$ satisfying (11), such that for any $t \in [0, T]$ and any $\delta \in (0, \min \{ T - t, e^{-e^{\max\{W(1/(2H)), 1\}}} \}]$,

$$|B^H(t+h) - B^H(t)| \leq C\delta^H \sqrt{\log \log \delta^{-1}}, \text{ a.s., for any } |h| \leq \delta, \quad (12)$$

where W denotes the Lambert W function: for every $z > 0$, $W(z)$ is the unique positive-valued solution of $xe^x = z$.

Proof. The function $x \mapsto x^H \sqrt{\log \log x^{-1}}$ is increasing over $(0, \min \{ e^{-e}, e^{-e^{W(1/(2H))}} \}]$.

- We only show the asymptotic lower bound for $H < 1/2$: there is $c > 0$ such that

$$\mathbb{P}(\tau_1 > T) \geq cT^{-(1-H)}(\log T)^{-1/(2H)}(\log_2 T)^{-1/(2H)}(\log_3 T)^{-1/(2H)}, \text{ as } T \rightarrow +\infty.$$

- The idea is to take advantages of the following relation (Tauberian theorem, Bingham, 1989):

$$I(T) := \mathbb{E}\left[\left(\int_0^T e^{B^H(u)} du\right)^{-1}\right] \approx \mathbb{E}\left[e^{-\sup_{t \in [0, T]} B^H(t)}\right] \approx \mathbb{P}(\tau_1 > T), \quad (13)$$

as $T \rightarrow +\infty$.

- 1 Take the random variable $u^* \in [0, 1]$ for which $\sup_{t \in [0, 1]} B^H(t) = B^H(u^*)$.
- 2 Pick $C \geq 1$ in the small increments of fBm theorem, then by the intermediate value, there is $C_0 \geq 1$ such that

$$C_0^{-H} \underbrace{\sqrt{\log_2(e^e C_0^e)}} = C^{-1}, \text{ a.s..}$$

it was C_0^γ in Aurzada (2011)

- 3 Let

$$\epsilon_0 = \min \left\{ (C_0 T (\log_2 T)^{1/(2H)})^{-1}, e^{-e^{\max\{W(1/(2H)), 1\}}} \right\}. \quad (14)$$

ϵ_0 satisfies:

- (1) $|B^H(t+h) - B^H(t)| \leq C \epsilon_0^H \sqrt{\log \log \epsilon_0^{-1}}$, a.s., for any $|h| \leq \epsilon_0$,
- (2) $\mu([u^* - \epsilon_0, u^* + \epsilon_0] \cap [0, 1]) \geq \epsilon_0$, a.s..
- (3) $T^H |B^H(u) - B^H(u^*)| \leq c$ as $T \rightarrow +\infty$, where $c > 0$ is deterministic.
- (4) $\epsilon_0^{-1} \leq C_0 T (\log_2 T)^{1/(2H)} + \tilde{c}_2$, a.s.

Using the features of ε_0 ,

$$\begin{aligned}
I(T) &\leq \mathbb{E} \left[\left(\int_{[u^* - \varepsilon_0, u^* + \varepsilon_0] \cap [0, 1]} e^{-T^H(B^H(u^*) - B^H(u))} T \, du \right)^{-1} e^{-T^H B^H(u^*)} \right] \\
&\leq \mathbb{E} \left[\varepsilon_0^{-1} e^{C T^H \varepsilon_0^H \sqrt{\log_2 \varepsilon_0^{-1}}} T^{-1} e^{-T^H B^H(u^*)} \right] \\
&\leq e^{\tilde{c}_1} T^{-1} \mathbb{E} \left[(C_0 T (\log_2 T)^{1/(2H)} + \tilde{c}_2) (e^{-T^H B^H(u^*)}) \right] \\
&\leq e^{\tilde{c}_1} T^{-1} \left(\mathbb{E} [(C_0 T (\log_2 T)^{1/(2H)} + \tilde{c}_2)^p] \right)^{1/p} \left(\mathbb{E} [e^{-q T^H B^H(u^*)}] \right)^{1/q} \\
&\leq 2e^{\tilde{c}_1} T^{-1} \left(\mathbb{E} [C_0^p T^p (\log_2 T)^{p/(2H)} + (\tilde{c}_2)^p] \right)^{1/p} \left(\mathbb{E} [e^{-q T^H B^H(u^*)}] \right)^{1/q} \\
&\leq c_1 (\mathbb{E} [C_0^p])^{1/p} (\log_2 T)^{1/(2H)} \left(\mathbb{E} [e^{-q T^H B^H(u^*)}] \right)^{1/q}, \text{ as } T \rightarrow +\infty,
\end{aligned} \tag{15}$$

Using the following inequalities

- ① $(\mathbb{E}[C_0^p])^{1/p} \leq c(\log \eta^{-1})^{1/(2(H-\eta))} p^{1/(2(H-\eta))}$ for $\eta \rightarrow 0^+$.
- ② $\mathbb{E}[e^{-qB^H(u^*)}] \leq \mathbb{E}[e^{-B^H(u^*)}]$ for $q > 1$.
- ③ (Molchan, 1999) $I(T) \geq cT^{-(1-H)}$, as $T \rightarrow +\infty$,

we obtain

$$\mathbb{E}[e^{-T^H B^H(u^*)}] \geq c_4^{-q} (\log \eta^{-1})^{-q/(2(H-\eta))} p^{-q/(2(H-\eta))} (\log_2 T)^{-q/(2H)} T^{-q(1-H)}. \quad (16)$$

Set $p = \log T$ and $\eta = (\log_2 T)^{-1}$. Note that $q = 1/(1 - 1/p) = 1 + \mathcal{O}(p^{-1})$ as $p \rightarrow +\infty$, and $1/(H - \eta) = 1/H + \mathcal{O}(\eta)$ as $\eta \rightarrow 0$. Then the right-hand side of (16) can be written as

$$\begin{aligned}
 & \exp \left\{ -q \left(\log(c_4) + \frac{\log_2 \eta^{-1}}{2(H - \eta)} + \frac{\log p}{2(H - \eta)} + \frac{\log_3 T}{2H} + (1 - H) \log T \right) \right\} \\
 &= \exp \left\{ - (1 + \mathcal{O}(p^{-1})) \left(\log(c_4) + \left(\frac{\log_4 T}{2H} + \mathcal{O}\left(\frac{\log_4 T}{\log_2 T}\right) \right) \right. \right. \\
 &\quad \left. \left. + \left(\frac{\log_2 T}{2H} + \mathcal{O}(1) \right) + \frac{\log_3 T}{2H} + (1 - H) \log T \right) \right\} \\
 &= \exp \left\{ - \left((1 - H) \log T + \frac{\log_2 T}{2H} + \frac{\log_3 T}{2H} + \frac{\log_4 T}{2H} + \mathcal{O}(1) \right) \right\},
 \end{aligned} \tag{17}$$

as $T \rightarrow +\infty$,

- We have proved that, there is a constant $c > 0$ such that

$$\mathbb{E}\left[e^{-T^H B^H(u^*)}\right] \geq c T^{-(1-H)} (\log T)^{-1/(2H)} (\log_2 T)^{-1/(2H)} (\log_3 T)^{-1/(2H)}, \quad (18)$$

as $T \rightarrow +\infty$.

- Tauberian theorem says that if $\{Y(u)\}_{u \in [0, +\infty)}$ is a self-similar stationary increment process,

$$\mathbb{P}\left(\sup_{u \in [0, 1]} Y(Tu) \leq 1\right) \sim \mathbb{E}\left[e^{\sup_{u \in [0, 1]} Y(Tu)}\right], \text{ as } T \rightarrow +\infty.$$

- As a result, for the fBm B^H we have: there exist $c_1, c_2 > 0$ such that

$$\begin{aligned} \mathbb{P}\left(\sup_{u \in [0, T]} B^H(u) \leq 1\right) &= \mathbb{P}\left(\sup_{u \in [0, 1]} B^H(Tu) \leq 1\right) \geq c_1 \mathbb{E}\left[e^{-B^H(Tu^*)}\right] \\ &= c_1 \mathbb{E}\left[e^{-T^H B^H(u^*)}\right] \geq c_2 T^{-(1-H)} ((\log T)(\log_2 T)(\log_3 T))^{-1/(2H)}, \end{aligned} \quad (19)$$

as $T \rightarrow +\infty$.

- Our dream is to prove the conjecture: for B^H ,

$$c_1 T^{-(1-H)} \leq \mathbb{P} \left(\sup_{t \in [0, T]} B^H(t) < b \right) \leq c_2 T^{-(1-H)}, \text{ as } T \rightarrow +\infty.$$

- We failed, but our efforts are not in vain.

In the future:

- Apply this methodology to the first exit time problem of other self-similar processes.

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